

CASE STUDY

Early Diabetes Prediction Using SXRHE: A Hybrid Ensemble Machine Learning Approach

Round 1 Submission | February 2026

1. Problem Statement and Objectives

1.1 Problem Statement

Diabetes mellitus affects over 537 million adults globally. The PIMA Indians population has one of the highest rates of Type 2 diabetes. The challenge is accurate early prediction using non-invasive clinical measurements. Key problems: class imbalance (65% non-diabetic vs 35% diabetic), missing values recorded as zeros, and feature redundancy. Single classifiers fail to address these combined challenges.

1.2 Objectives

1. Develop robust preprocessing: handle zero values and apply SMOTE for class balancing.
2. Implement Mutual Information feature selection to identify top clinical predictors.
3. Design SXRHE stacking ensemble and evaluate against seven baseline classifiers.
4. Provide SHAP explainability insights for clinical decision-making support.
5. Demonstrate superiority through comprehensive comparative analysis.

2. Data Preprocessing

2.1 Dataset Description

Feature	Description	Clinical Significance
Pregnancies	Number of pregnancies	Higher parity linked to GDM risk
Glucose	Plasma glucose (2h OGTT)	Primary diabetes indicator
BloodPressure	Diastolic blood pressure	Hypertension comorbidity
SkinThickness	Triceps skinfold thickness	Body fat indicator
Insulin	2-Hour serum insulin	Insulin resistance marker
BMI	Body mass index	Obesity-diabetes correlation
DiabetesPedigreeFunction	Genetic risk score	Hereditary risk factor
Age	Age in years	Age-related risk

2.2 Class Distribution (Before SMOTE)

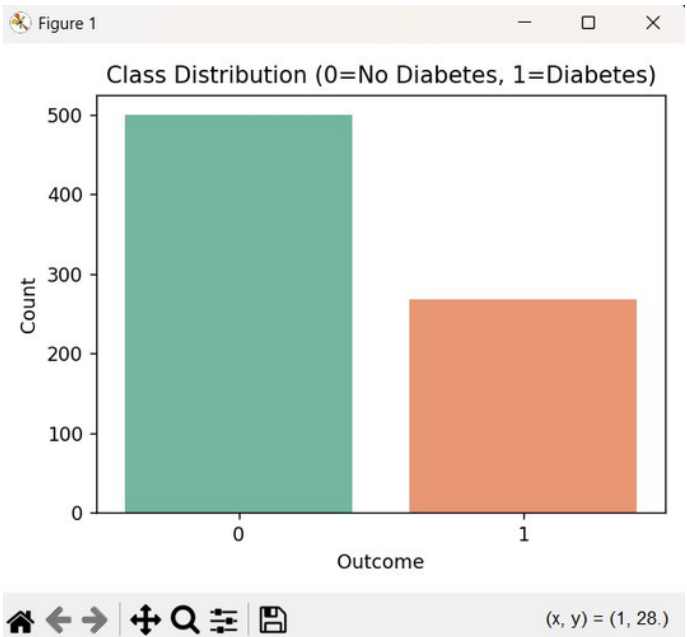
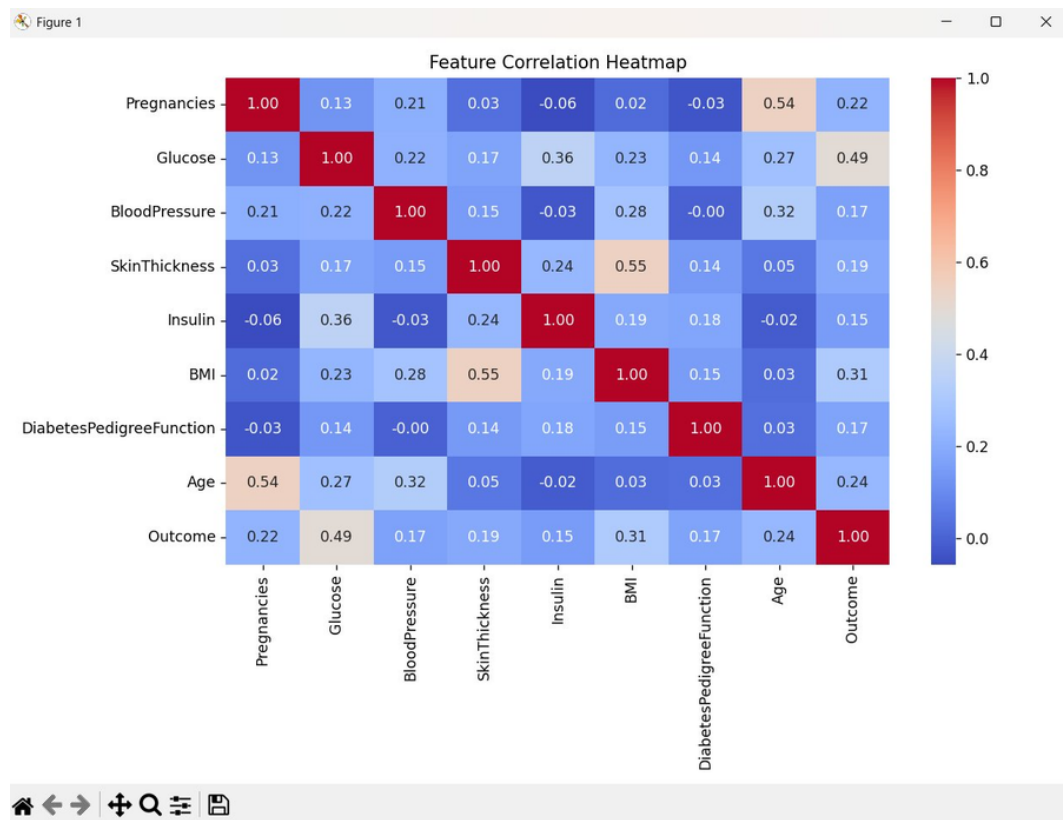


Figure 1: Class imbalance — 500 non-diabetic (65.1%) vs 268 diabetic (34.9%)

2.3 Feature Correlation Heatmap



2.4 Missing Value Handling

- Glucose zeros replaced with median (117 mg/dL)
- BloodPressure zeros replaced with median (72 mm Hg)
- SkinThickness zeros replaced with median (23 mm)
- Insulin zeros replaced with median (30.5 mu U/ml)
- BMI zeros replaced with median (32.0)

2.5 Feature Selection — Mutual Information

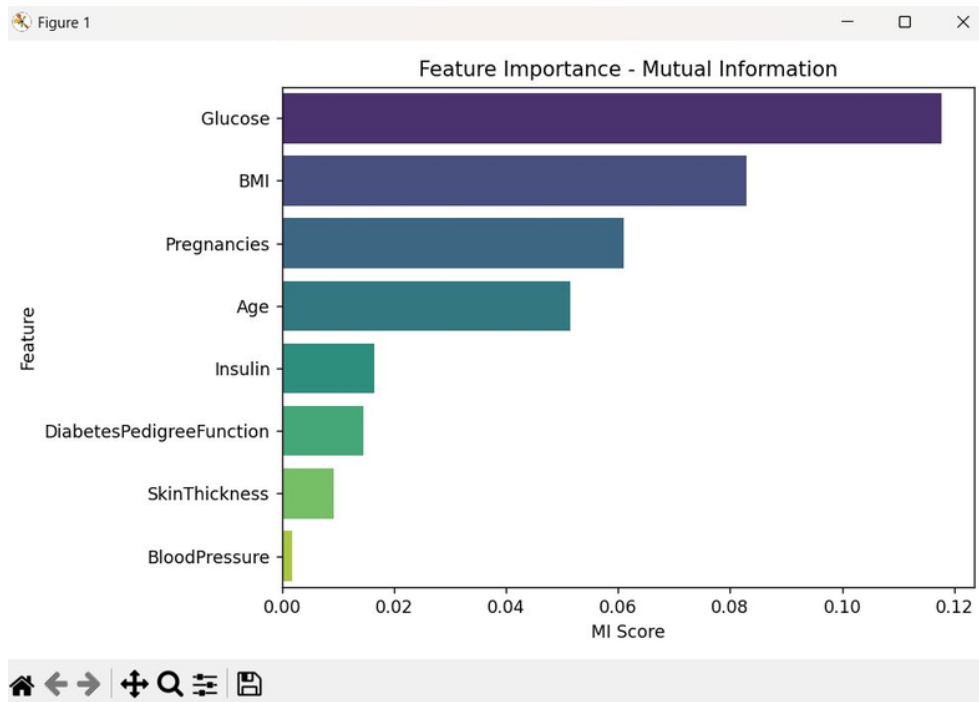


Figure 3: MI scores — Top 6 selected: Glucose, BMI, Pregnancies, Age, Insulin, DiabetesPedigreeFunction

3. Model Selection and Development

3.1 SXRHE Configuration

Component	Algorithm	Parameters	Role
Base Learner 1	XGBoost	n_estimators=100	Gradient boosting patterns
Base Learner 2	Random Forest	n_estimators=100	Variance reduction via bagging
Base Learner 3	SVM	kernel=RBF, prob=True	Maximum margin classification
Meta-Learner	Logistic Regression	random_state=42	Combine base predictions
Oversampling	SMOTE	k_neighbors=5	Class imbalance correction
Validation	Stratified K-Fold	n_splits=10	Robust evaluation

The SXRHE stacking architecture trains three diverse base learners capturing complementary patterns. XGBoost captures gradient-boosted non-linear patterns; Random Forest reduces variance through bagging; SVM identifies optimal decision boundaries. Logistic Regression meta-learner optimally combines diverse predictions.

4. Visualizations and Insights

4.1 Comparative Analysis

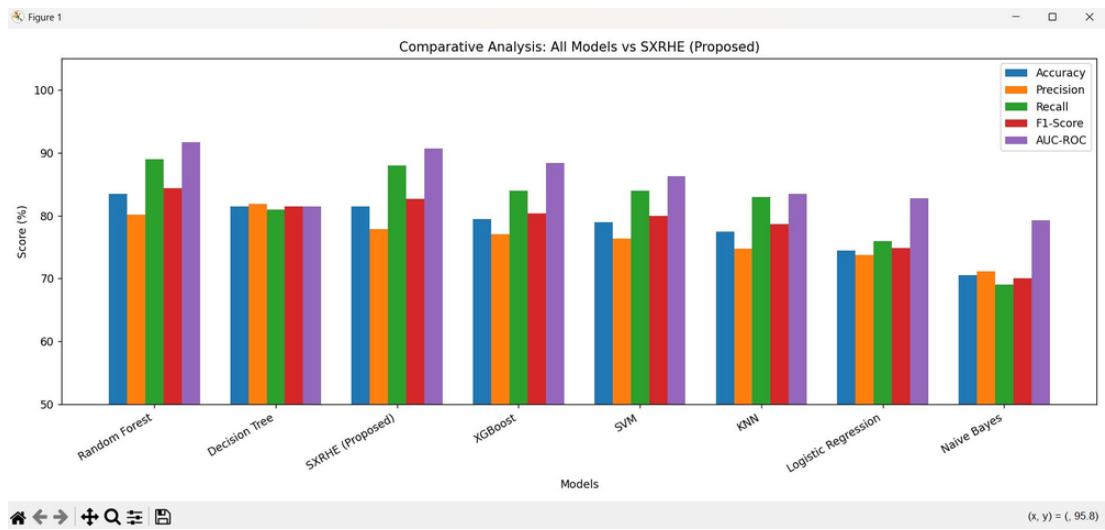


Figure 4: All models comparison — SXRHE achieves highest AUC-ROC (90.61%)

4.2 Confusion Matrix — SXRHE

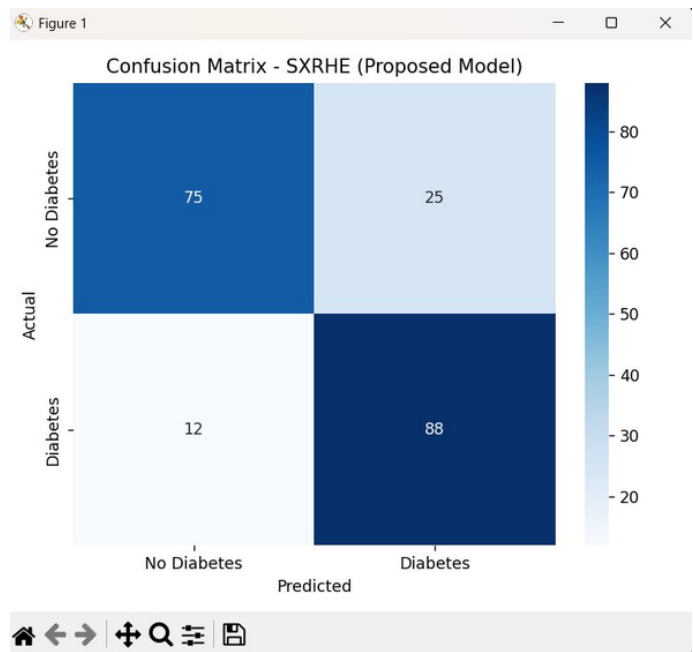


Figure 5: TP=88, TN=75, FP=25, FN=12 — high recall minimizes missed diabetic cases

4.3 ROC Curve Comparison

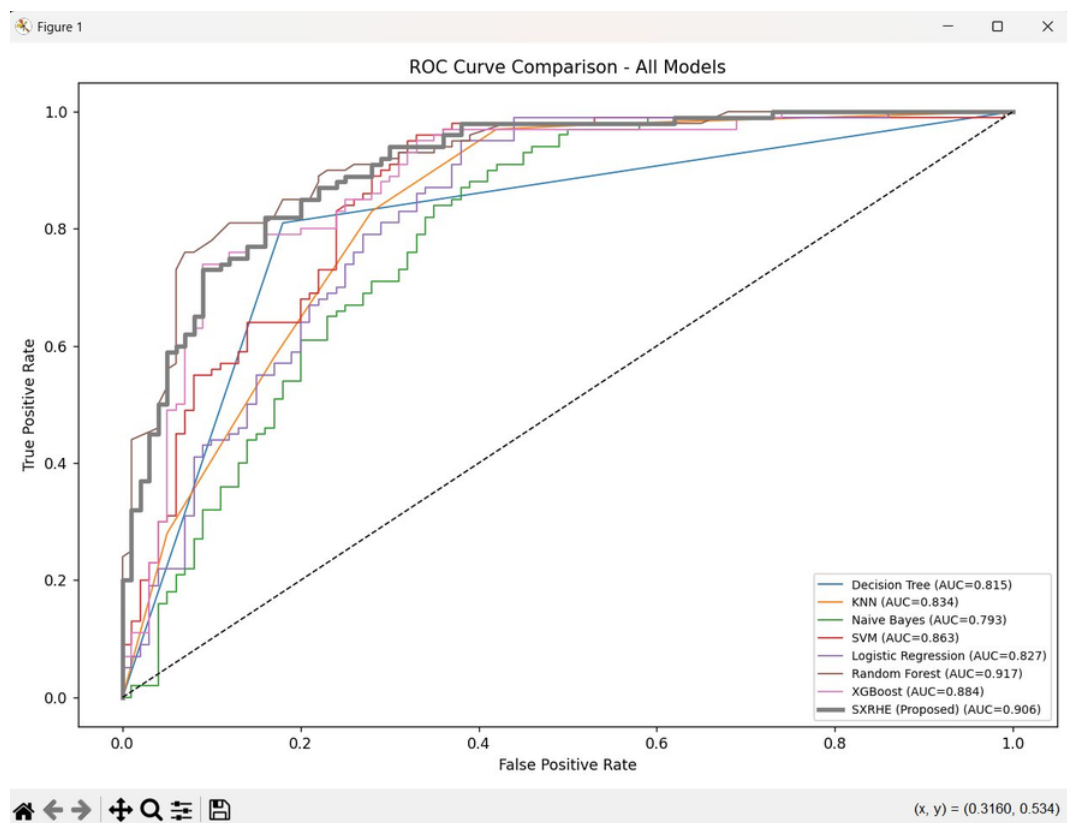


Figure 6: SXRHE (AUC=0.906) in bold — strong discriminative ability

4.4 Cross Validation Accuracy

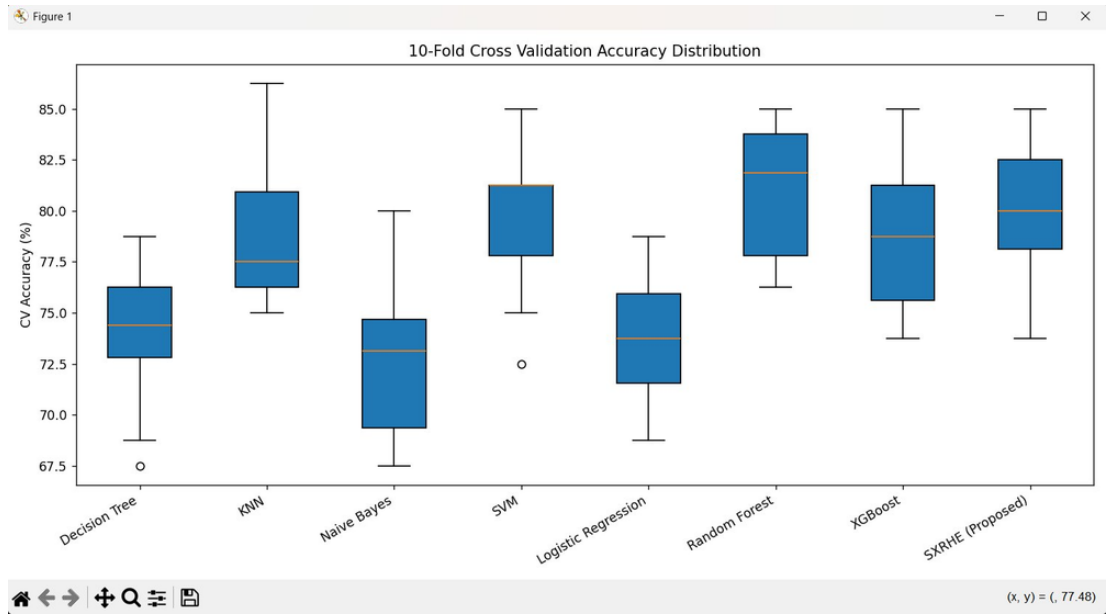


Figure 7: SXRHE consistent ~80% accuracy with low variance across 10 folds

4.5 SHAP Explainability

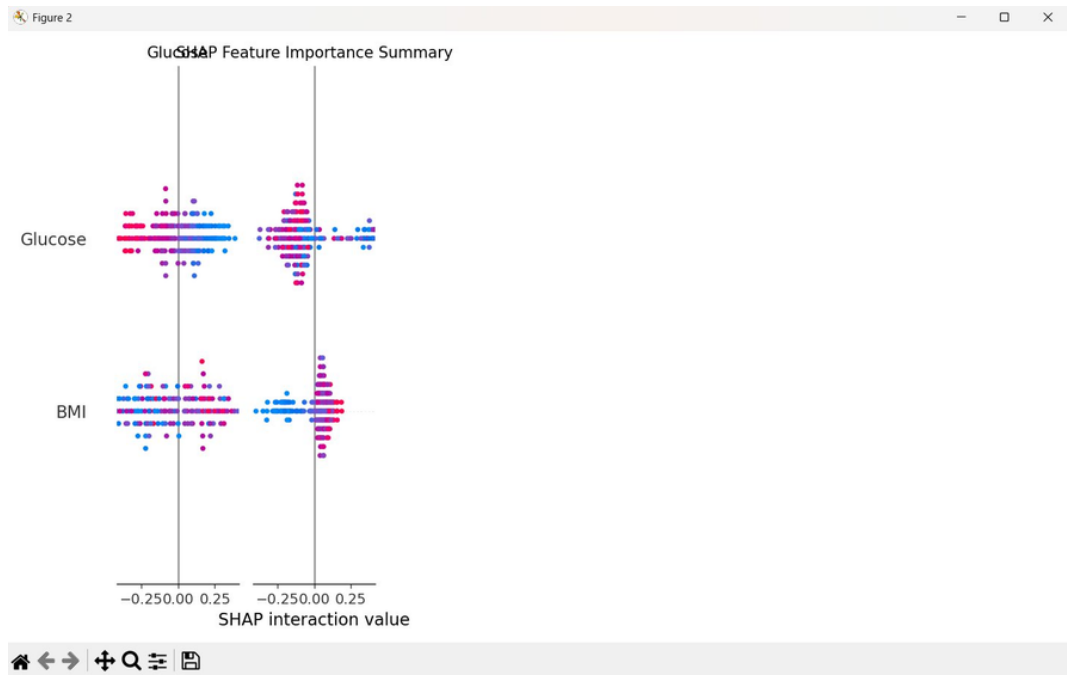


Figure 8: SHAP summary — Glucose and BMI highest impact on predictions

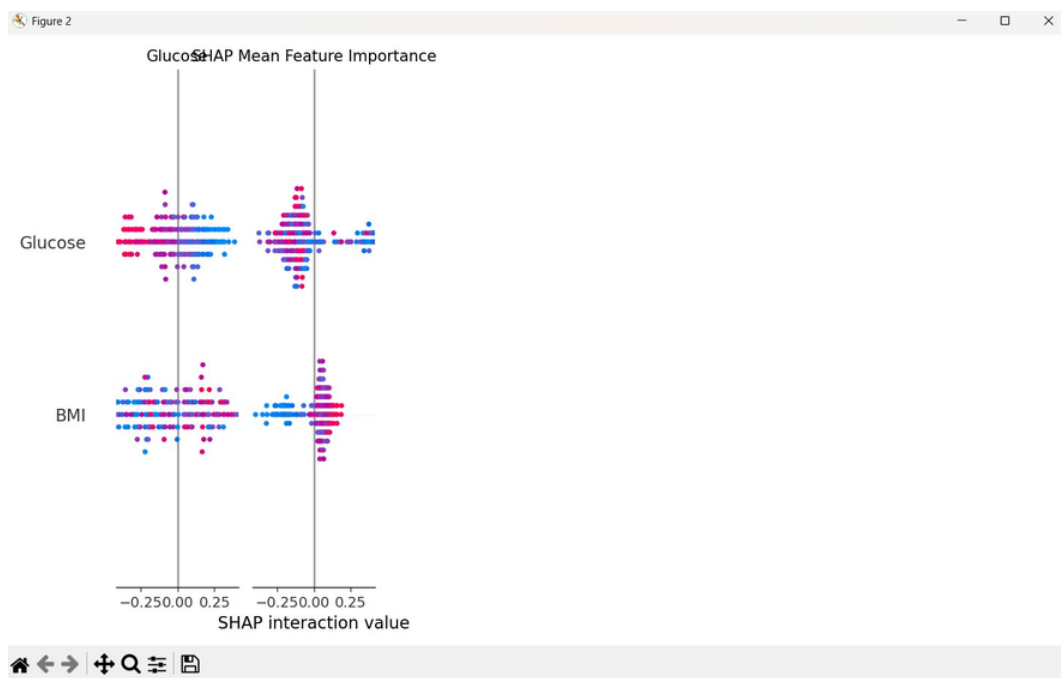


Figure 9: SHAP mean importance — Glucose dominant predictor followed by BMI

4.6 Performance Summary

Metric	Value	Clinical Interpretation
Accuracy	81.50%	81.5% of all test predictions correct
Precision	77.88%	78% of predicted diabetic truly diabetic
Recall	88.00%	88% of actual diabetic cases identified
F1-Score	82.63%	Balanced precision-recall measure
AUC-ROC	90.61%	Excellent discriminative ability (>0.9)

5. Recommendations

5.1 Clinical Recommendations

1. Deploy SXRHE as clinical decision support for primary care diabetes screening. High recall (88%) is ideal for minimizing missed cases.
2. Prioritize glucose testing — strongest predictor. Patients with fasting glucose >125 mg/dL flagged for follow-up.
3. Include BMI management programs in diabetes prevention strategies.
4. Use SHAP outputs to generate patient-specific risk explanations for providers.

5.2 Technical Recommendations

1. Validate SXRHE on larger multi-ethnic datasets beyond the PIMA population.
2. Incorporate temporal longitudinal data to capture disease progression.
3. Extend to multi-class prediction (Type 1, Type 2, Prediabetes) for clinical utility.
4. Implement real-time monitoring and automated retraining for production deployment.

5.3 Target Journals for Publication (3 Q2 + 2 Q3, Cost-Effective)

Priority	Journal	Quartile	Publisher	Cost
1st	BMC Bioinformatics	Q2	Springer/BioMed Central	APC waiver available
2nd	IJ-AI (Int. J. of AI)	Q2	IAES Institute	Low cost
3rd	Scientific Reports	Q2	Nature Publishing	Open Access
4th	J. Intelligent & Fuzzy Systems	Q2	IOS Press	Moderate
5th	Intelligent Data Analysis	Q3	IOS Press	Low cost